Verbetering in het groen

Machines & Berekenbaarheid - Oefeningenreeks 1

① a) $(V = \{V\}, T = \{0,1\}, P = \{V \to 01 \mid 0V1\}, \cancel{S = V\}}$
$S = V)$

OK

b) (V, T, P, S)

$V = \{A, B, C\}$
$T = \{x, y\}$
$S = A$

ε | ε
xy of xyy
$xxyy$ | $xxyyyy$

$\cancel{V_1 \to \varepsilon \mid xy \mid xV_1y \mid xV_2yy}$

$P =$
Eerst kiezen:
$A \to \varepsilon \mid B \mid C$
Dan bij die keuze blijven:
$B \to xy \mid xBy$
$C \to xyy \mid xCyy$

OK

c) zzzz | zzzz
zazazz en zzbzbz
zaazaazz | zzbbzbbz

$V = \{A, B, C\}$
$T = \{z, a, b\}$
$S = A$

$P = \{A \to zBCz$
$B \to z \mid aBa$
$C \to z \mid bCb\}$

OK

~~d)~~
$\cancel{V = \{A, B, C, D, E, F\} \quad P =}$
$\cancel{T = \{a, b, c\}}$
$\cancel{S = A}$

$P = \{A \to BC \mid CD, B \to aB \mid \varepsilon, \cancel{C \to \ldots \mid c}$

d) Geen idee Examenvraag

$G = \{\{S, A, B, C, D, E\}, \{a, b, c\}, P, S\}$

$P = \{ S \to AB \mid CD, \quad B \to cB \mid \varepsilon, \quad D \to bDc \mid bE \mid cB$
$A \to aAb \mid aC \mid bE, \quad C \to aC \mid \varepsilon, \quad E \to bE \mid \varepsilon \}$

② ~~ab~~ Geen examenvraag

Linker: $S \xRightarrow{1} A1B \xRightarrow{2} 0A1B \xRightarrow{2} 00A1B \xRightarrow{3} 001B \xRightarrow{4} 0010B$
$\xRightarrow{5} 00101B \xRightarrow{6} 00101$

Rechter: $S \xRightarrow{1} A1B \xRightarrow{4} A10B \xRightarrow{5} A101B \xRightarrow{6} A101 \xRightarrow{2} 0A101$
$\xRightarrow{2} 00A101 \xRightarrow{3} 00101$ OK

③ (V, T, P, S)

$V = \{A, B, C, D\}$

$T = \{a, b, c\}$

$S = \cancel{A}\ B$

$$P = \left\{ \begin{array}{l} \cancel{A}B \rightarrow BCbA \\ \cancel{B}C \rightarrow \varepsilon \mid aC \\ D \rightarrow \varepsilon \mid bD \mid cD \\ \cancel{B}A \rightarrow \varepsilon \mid bA \end{array} \right\}$$

OK

④ (V, T, P, S)

$T = \{0, 1, (,), +, \cdot, \emptyset, e\}$

$S = L$

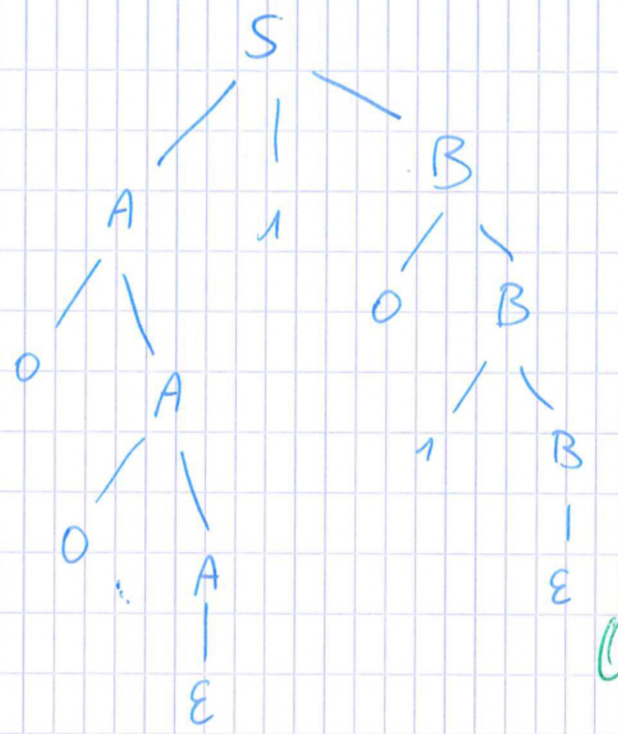
$V = \{S, L, E, I$

$$P = \left\{ \begin{array}{l} L \rightarrow \emptyset \\ L \rightarrow E \\ E \rightarrow E + E \\ E \rightarrow (E)^* \\ E \rightarrow I \\ I \rightarrow 0 \mid 1 \mid I0 \mid I1 \mid e \\ S \rightarrow L \mid E \mid I \end{array} \right\}$$

Machines & Berekenbaarheid – Oefeningenreeks 2

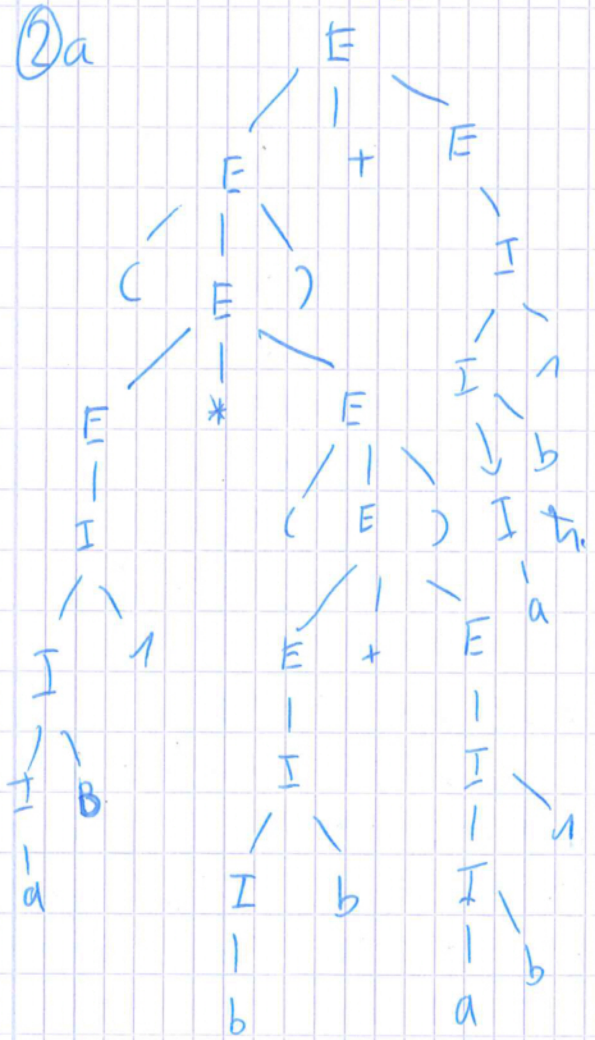
Verbetering in het groen

1



OK

2a



OK

Geen examenvraag

b) $E \overset{2}{\Rightarrow} E+E \overset{1}{\Rightarrow} E+I \overset{10}{\Rightarrow} E+E1 \overset{8}{\Rightarrow} E+Ib1 \overset{5}{\Rightarrow} E+ab1$
$\overset{8}{\Rightarrow} (E)+ab1 \overset{3}{\Rightarrow} (E*E)+ab1 \overset{4}{\Rightarrow} (E*(E))+ab1$
$\overset{2}{\Rightarrow} (E*(E+E))+ab1 \overset{1}{\Rightarrow} (E*(E+I))+ab1$
$\overset{10}{\Rightarrow} (E*(E+I1))+ab1 \overset{8}{\Rightarrow} (E*(E+Ib1))+ab1$
$\overset{5}{\Rightarrow} (E*(E+ab1))+ab1 \overset{1}{\Rightarrow} (E*(I+ab1))+ab1$
$\overset{8}{\Rightarrow} (E*(Ib+ab1))+ab1 \overset{6}{\Rightarrow} (E*(bb+ab1))+ab1$
$\overset{1}{\Rightarrow} (I*(bb+ab1))+ab1 \overset{10}{\Rightarrow} (I1*(bb+ab1))+ab1$
$\overset{8}{\Rightarrow} (Ib1*(bb+ab1))+ab1 \overset{5}{\Rightarrow} (ab1*(bb+ab1))+ab1$

OK

c)

inferred strings	language	productions	strings
a	I	5	/
ab	I	8	1
ab1	I	10	2
b	I	6	/
bb	I	8	4
ab1	E	1	3
bb	E	1	5
bb + ab1	E	2	6, 7
(bb + ab1)	E	4	8
ab1 + (bb + ab1)	E	2	6, 9
(ab1 + (bb + ab1))	E	4	10
(ab1 + (bb + ab1)) + ab1	E	2	6, 11

Geen examenvraag

OK

3 a) $((aa*bb)+a1b)*(a*(b0+b))$ OK

b) $E \overset{3}{\Rightarrow} E*E \overset{4}{\Rightarrow} (E)*E \overset{2}{\Rightarrow} (E+E)*E \overset{4}{\Rightarrow} ((E)+E)*E \overset{3}{\Rightarrow} ((E*E)+E)*E$
$\overset{1}{\Rightarrow} ((I*E)+E)*E \overset{7,5}{\Rightarrow} ((aa*E)+E)*E \overset{1,8,6}{\Rightarrow} ((aa*bb)+E)*E$
$\overset{1,8,10,5}{\Rightarrow} ((aa*bb)+a1b)*E \overset{4,1,3,5}{\Rightarrow} ((aa*bb)+a1b)*(a*E)$
$\overset{1,2,4,6,9}{\Rightarrow} ((aa*bb)+a1b)*(a*(b0+E))$
$\overset{1,6}{\Rightarrow} ((aa*bb)+a1b)*(a*(b0+b))$

Geen examenvraag

OK

Verbetering in het groen

Machines & Berekenbaarheid - Oefeningenreeks 3

① Inductiebewijs:

Basisgeval:

- $|w| = 0 \Rightarrow w = \varepsilon \;||\; B \Rightarrow \cancel{()}\ \varepsilon$
- $|w| = 2 \Rightarrow w = () \;||\; B \Rightarrow (B) \Rightarrow ()$

IH: Als $|w| = 2n$ ~~met~~ w en $\exists n \in \mathbb{N}$

Inductiebewijs:

$|w| = 2n + 2$

- $w = (w_1) \leftarrow IH$
  en $w_1 \in W$
- $w = w_1 w_2$ met $w_1, w_2 \notin \varepsilon$
  $B = BB \overset{IH}{\Rightarrow} w_1 w_2$

~~QED~~ □ OK

② (V, T, P, S)

$V = B$

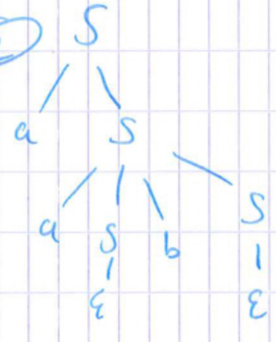
$T = \{ (,), [,] \}$

$P = \{ B \to \varepsilon \mid BB \mid [B] \mid (B) \}$

$S = B$ OK

③ ①

②



OK

b) $S = aSbS = aaSBs = aabS = aab$ ① OK

$S = aS = aaSbS = aabS = aab$ ② OK

c) $S = aSbS = aSb = aaSb = aab$ ① OK

$S = aS = aaSbS = aaSb = aab$ ② OK